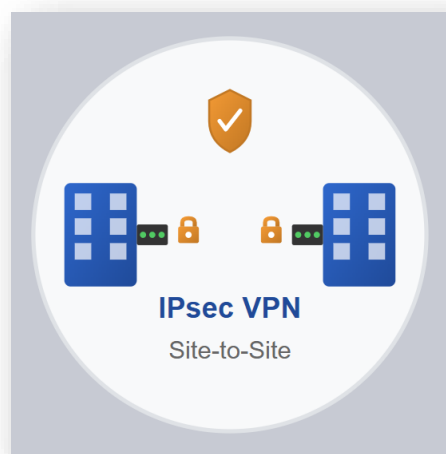


Cycle ingénieur – Intelligence Artificielle et Cybersécurité

Module : Sécurité des Réseaux et Protocoles

PROJET : *Configuration d'un VPN Site-à-Site IPsec avec Cisco IOS*



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INTRODUCTION

Ce TP a pour objectif de configurer un VPN site-à-site sécurisé entre deux réseaux distants en utilisant le protocole IPsec sur des routeurs Cisco IOS. Les routeurs R1 et R3 représentent deux sites distants, tandis que R2 joue le rôle de fournisseur d'accès (FAI) sans connaissance du VPN.

Un dépôt GitHub est créé pour centraliser la documentation, les configurations et les ressources du projet.

Lien du dépôt GitHub : [BOUBKERsup07/Lab_configure-site-to-site-vpn-using-cli](https://github.com/BOUBKERsup07/Lab_configure-site-to-site-vpn-using-cli)

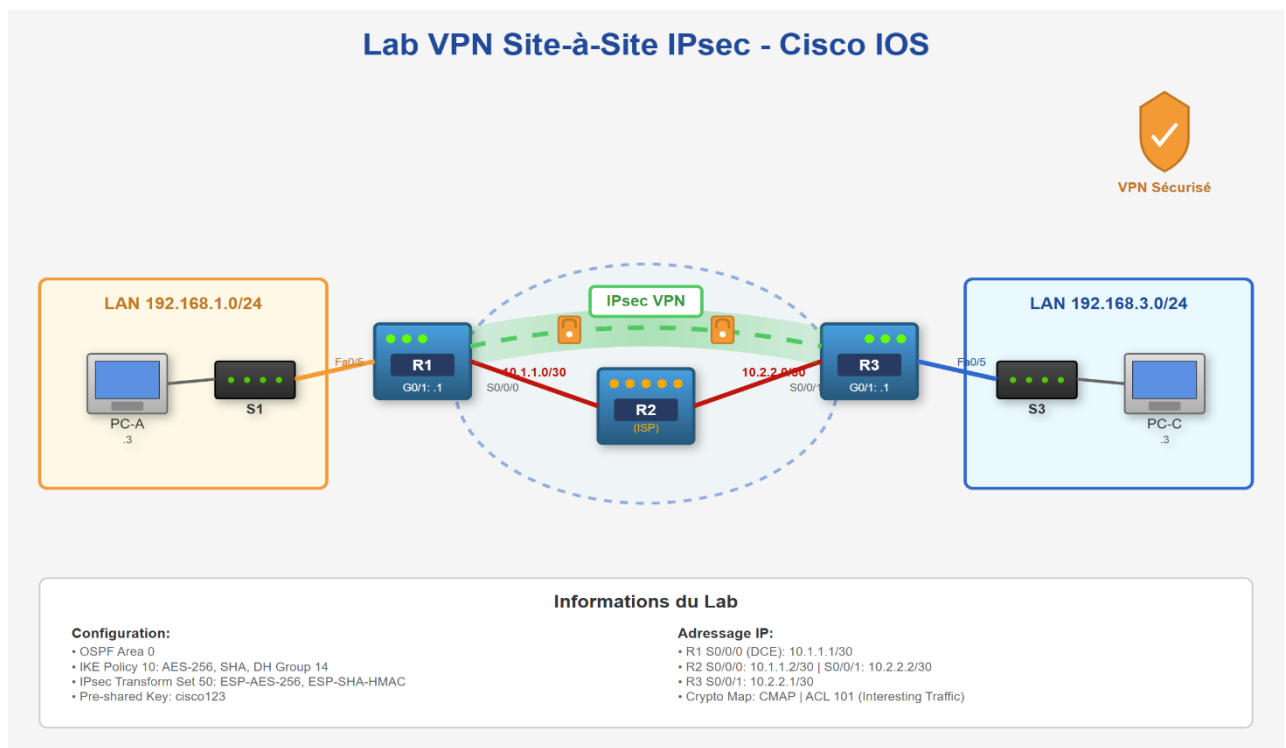


Figure 1- Lab VPN IPsec - Architecture Trois Routeurs

Parti 1 : Configure Basic Device Settings

Step 1: Cable the network as shown in the topology.

Table d'adressage IP

<i>Routeur</i>	<i>Interface</i>	<i>Adresse IP</i>	<i>Masque</i>
<i>R1</i>	FastEthernet0/0	192.168.1.1	255.255.255.0
<i>R1</i>	Serial2/0	10.1.1.1	255.255.255.252
<i>R2</i>	Serial2/0	10.1.1.2	255.255.255.252
<i>R2</i>	Serial3/0	10.2.2.2	255.255.255.252
<i>R3</i>	FastEthernet0/0	192.168.3.1	255.255.255.0
<i>R3</i>	Serial3/0	10.2.2.1	255.255.255.252

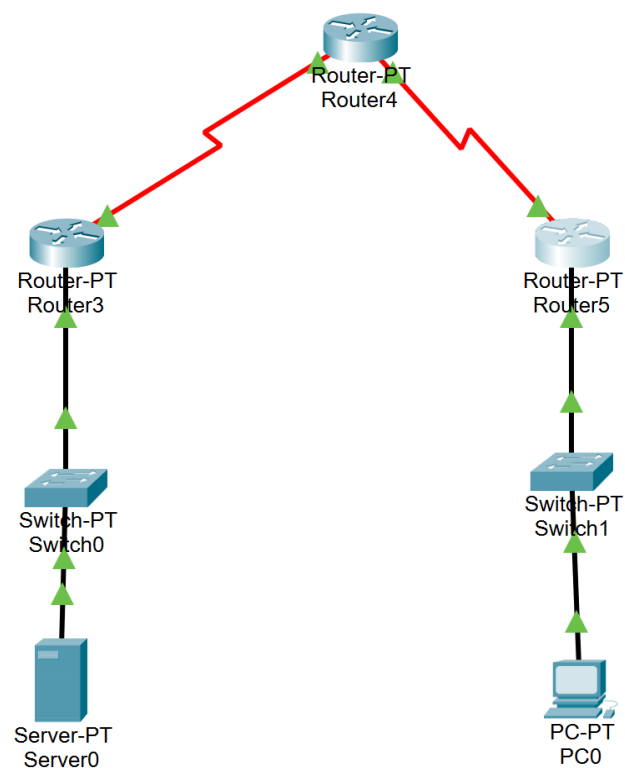


Figure 2: Vue globale de la topologie Packet Tracer

Step 2: Configure basic settings for each router.

- a. *Configure hostnames, as shown in the topology.*

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show running-config | include hostname
hostname R1
R1#
```

Figure 3 - Configure hostnames R1

```
R2>enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#hostname R2
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show running-config | include hostname
hostname R2
R2#
```

Figure 4 - Configure hostnames R2

```
R3>enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#hostname R3
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show running-config | include hostname
hostname R3
R3#
```

Figure 5 - Configure hostnames R3

- b. *Configure the interface IP addresses, as shown in the IP Addressing Table.*

Commande :

```
show ip interface brief
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
```

```
R1#show ip interface brief
Interface          IP-Address      OK? Method Status              Protocol
FastEthernet0/0    192.168.1.1     YES manual up                  up
FastEthernet1/0    unassigned      YES unset administratively down down
Serial2/0          10.1.1.1        YES manual up                  up
Serial3/0          unassigned      YES unset administratively down down
FastEthernet4/0    unassigned      YES unset administratively down down
FastEthernet5/0    unassigned      YES unset administratively down down
R1#
```

Figure 6 - ip interface R1

```
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface Serial2/0
R2(config-if)# ip address 10.1.1.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)#exit
R2(config)#
```

```
R2#show ip interface brief
Interface          IP-Address      OK? Method Status              Protocol
FastEthernet0/0    unassigned      YES unset administratively down down
FastEthernet1/0    unassigned      YES unset administratively down down
Serial2/0          10.1.1.2        YES manual up                  up
Serial3/0          10.2.2.2        YES manual down                down
FastEthernet4/0    unassigned      YES unset administratively down down
FastEthernet5/0    unassigned      YES unset administratively down down
R2#
```

Figure 7 - ip interface R2

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)#exit
R3(config)#
```



```
R3#enable
R3#show ip interface brief
Interface                IP-Address      OK? Method Status              Protocol
FastEthernet0/0          192.168.3.1     YES manual up                  up
FastEthernet1/0          unassigned      YES unset  administratively down down
Serial2/0                 unassigned      YES unset  administratively down down
Serial3/0                 10.2.2.1        YES manual down                  down
FastEthernet4/0          unassigned      YES unset  administratively down down
FastEthernet5/0          unassigned      YES unset  administratively down down
R3#
```

Figure 8 - ip interface R3

- c. Configure a clock rate of 64000 for the serial router interfaces with a DCE serial cable attached.

```
R3(config)#interface Serial3/0
R3(config-if)# ip address 10.2.2.1 255.255.255.252
R3(config-if)# clock rate 64000
R3(config-if)# no shutdown
R3(config-if)#exit
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

Comment :

show running-config | section interface Serial

```
R3#show running-config | section interface Serial
interface Serial2/0
  no ip address
  clock rate 2000000
  shutdown
interface Serial3/0
  ip address 10.2.2.1 255.255.255.252
  clock rate 64000
R3#
```

Figure 9 - Configure clock

Step 3: Disable DNS lookup.

Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands.

```
R1(config)#no ip domain-lookup
```

Commend :

show running-config | include domain-lookup

```
R1#show running-config | include domain-lookup
no ip domain-lookup
R1#
```

```
R2#show running-config | include domain-lookup
no ip domain-lookup
R2#
```

```
R3#show running-config | include domain-lookup
no ip domain-lookup
R3#
```

Figure 10 - Disable Dns lookup

Step 4: Configure the OSPF routing protocol on R1, R2, and R3.

a. **On R1, use the following commands:**

Commend : show ip route

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 101
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# network 10.1.1.0 0.0.0.3 area 0
R1(config-router)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      10.0.0.0/30 is subnetted, 1 subnets
C       10.1.1.0 is directly connected, Serial2/0
C      192.168.1.0/24 is directly connected, FastEthernet0/0

R1#
```

Figure 11 - Configure the OSPF R1

b. On R2, use the following commands:

Commend : show ip route

```
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router ospf 101
R2(config-router)# network 10.1.1.0 0.0.0.3 area 0
R2(config-router)# network 10.2.2.0 0.0.0.3 area 0
R2(config-router)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 1 subnets
C       10.1.1.0 is directly connected, Serial2/0
O       192.168.1.0/24 [110/65] via 10.1.1.1, 01:32:26, Serial2/0

R2#
```

Figure 12 - Configure the OSPF R2

c. On R3, use the following commands:

Commend : show ip route

```
R3>enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 101
R3(config-router)# network 192.168.3.0 0.0.0.255 area 0
R3(config-router)# network 10.2.2.0 0.0.0.3 area 0
R3(config-router)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.3.0/24 is directly connected, FastEthernet0/0
```

Figure 13 - Configure the OSPF R3

Step 5: Configure PC host IP settings.

- Configure a static IP address, subnet mask, and default gateway for PC-A, as shown in the IP Addressing Table.

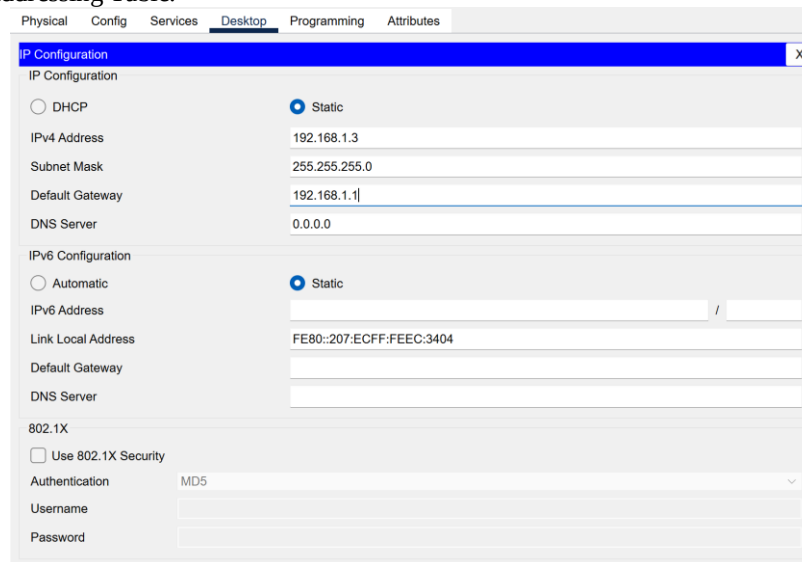


Figure 14 - Configure PC-A

- Configure a static IP address, subnet mask, and default gateway for PC-C, as shown in the IP Addressing Table.

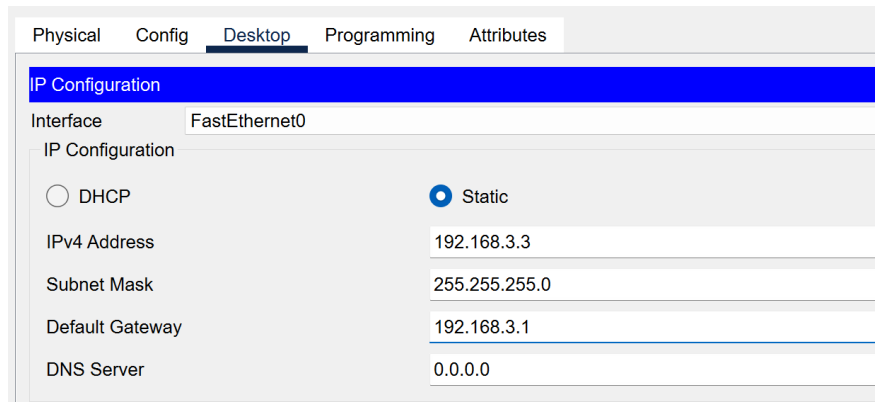


Figure 15 - Configure PC-C

Step 6: Verify basic network connectivity.

- a. Ping from R1 to the R3 Fa0/1 interface at IP address 192.168.3.1.

```
R1#ping 192.168.3.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 37/47/53 ms
```

```
R1#
```

Figure 16 - Ping from R1 to the R3 G0/1 interface.

- b. Ping from PC-A on the R1 LAN to PC-C on the R3 LAN.

```
C:\>ping 192.168.3.3
```

```
Pinging 192.168.3.3 with 32 bytes of data:
```

```
Reply from 192.168.3.3: bytes=32 time=62ms TTL=125
```

```
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125
```

```
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125
```

```
Reply from 192.168.3.3: bytes=32 time=74ms TTL=125
```

```
Ping statistics for 192.168.3.3:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 2ms, Maximum = 74ms, Average = 35ms
```

```
C:\>
```

Figure 17 - Ping from PC-A on the R1 LAN to PC-C on the R3 LAN.

Step 7: Configure and encrypt passwords.

Pour R1:

- a. Configure a minimum password length.

Use the **security passwords** command to set a minimum password length of 10 characters

Commande non supportée par la version IOS utilisée.

```
R1>enable
R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#security passwords min-length 10
      ^
% Invalid input detected at '^' marker.
```

Figure 18 - Configure a minimum password length.

- b. Configure the enable secret password on both routers with a password of **cisco12345**. Use the type 9 (SCRYPT) hashing algorithm.

show running-config | include enable secret

```
R1(config)#enable secret cisco12345
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show running-config | include enable secret
enable secret 5 $1$mERr$WvpW0n5HghRrqnrxXCUU1.
R1#
```

Figure 19 - Enable secret Pour R1

- c. Create a local **admin01** account using **admin01pass** for the password. Use the type 9 (SCRYPT) hashing algorithm.

show running-config | section username

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#username admin01 secret admin01pass
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show running-config | section username
username admin01 secret 5 $1$mERr$wm5onR0mqpKAz86A1LV8W1
R1#
```

Figure 20 - Compte local admin01 Pour R1

Pour R3:

```
R3>
R3>enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#enable secret cisco12345
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show running-config | include enable secret
enable secret 5 $1$mERr$WvpW0n5HghRrqn timerwXCUU1.
R3#
```

Figure 21- Enable secret Pour R2

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#username admin01 secret admin01pass
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show running-config | section username
username admin01 secret 5 $1$mERr$wm5onR0mqpKAz86A1LV8W1
R3#
```

Figure 22- Compte local admin01 Pour R2

Step 8: Configure the console line.

Configure the console to use the local database for login. For additional security, configure the line to log out after five minutes of inactivity. Issue the **logging synchronous** command to prevent console messages from interrupting command entry.

Pour R1:

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#line console 0
R1(config-line)# login local
R1(config-line)# exec-timeout 5 0
R1(config-line)# logging synchronous
R1(config-line)#exit
R1(config)#end
R1#
```

show running-config

```
!  
line con 0  
  exec-timeout 5 0  
  logging synchronous  
  login local  
!
```

Figure 23 – Configuration de la console R1

Pour R3:

```
R3#configure terminal  
Enter configuration commands, one per line.  End with CNTL/Z.  
R3(config)#line console 0  
R3(config-line)# login local  
R3(config-line)# exec-timeout 5 0  
R3(config-line)# logging synchronous  
R3(config-line)#exit  
R3(config)#end  
R3#
```

show running-config

```
!  
line con 0  
  exec-timeout 5 0  
  logging synchronous  
  login local  
!
```

Figure 24 – Configuration de la console R3

Step 9: Configure SSH Server.

Pour R1:

- a. Configure a domain name **ccnasecurity.com**.

show running-config | include domain-name


```
R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip domain-name ccnasecurity.com
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show running-config | include domain-name
ip domain-name ccnasecurity.com
R1#
```

Figure 25 - Configuration domaine R1

- b. Configure the RSA keys with **1024** for the number of modulus bits.

Message de génération des clés RSA

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto key generate rsa general-keys modulus 1024
The name for the keys will be: R1.ccnasecurity.com

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 4:41:16.735: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
```

Figure 26 - Génération des clés RSA R1

- c. Issue the command to force the use of SSH version 2.

show ip ssh

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip ssh version 2
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#
```

Figure 27 - Forcer l'utilisation de SSH version 2 R1

- d. Configure the vty lines on R1 and R3 to use the local database for login. Remote access to the routers should only be allowed using SSH. Configure the vty lines to logout after five minutes of inactivity.

show running-config | section line vty

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#exec-timeout 5 0
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show running-config | section line vty
line vty 0 4
  exec-timeout 5 0
  login local
  transport input ssh
R1#
```

Figure 28 - Configuration des lignes VTY R1

Pour R3:

- a. Configure a domain name **ccnasecurity.com**.

show running-config | include domain-name

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip domain-name ccnasecurity.com
R3(config)#end
R3#

R3#show running-config | include domain-name
ip domain-name ccnasecurity.com
R3#
```

Figure 29 - Configuration domaine R3

- b. Configure the RSA keys with **1024** for the number of modulus bits.

Message de génération des clés RSA

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#crypto key generate rsa general-keys modulus 1024
The name for the keys will be: R3.ccnasecurity.com

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 4:56:8.937: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 30 - Génération des clés RSA R3

- c. Issue the command to force the use of SSH version 2.

show ip ssh

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip ssh version 2
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
n34l
```

Figure 31 - Forcer l'utilisation de SSH version 2 R3

- d. Configure the vty lines on R1 and R3 to use the local database for login. Remote access to the routers should only be allowed using SSH. Configure the vty lines to logout after five minutes of inactivity.

show running-config | section line vty

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#exec-timeout 5 0
R3(config-line)#transport input ssh
R3(config-line)#exit
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show running-config | section line vty
line vty 0 4
  exec-timeout 5 0
  login local
  transport input ssh
R3#
```

Figure 32 - Configuration des lignes VTY R3

Step 10: Save the basic running configuration for all three routers.

Save the running configuration to the startup configuration from the privileged EXEC mode prompt on R1, R2, and R3.

copy running-config startup-config sur (R1,R2,R3)

```
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#enable
R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#

R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R3#
```

show startup-config

POUR R1:

```

!#show startup-config
Using 1158 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
enable secret 5 $1$mERr$WpW0n5HghRqnrwXCUU1.
!
!
!
ip cef
no ipv6 cef
!
!
username admin01 secret 5 $1$mERr$wm5onRmqpKAz8GillV8Wl
!
!
!
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name ccnasecurity.com
!
!
!
!
!
!
!
line con 0
exec-timeout 5 0
logging synchronous
login local
!
line aux 0
!
line vty 0 4
exec-timeout 5 0
login local
transport input ssh
!
interface Serial2/0
ip address 10.1.1.1 255.255.255.252
!
interface Serial3/0
no ip address
end

```

Figure 33 - Sauvegarde de la configuration R1

POUR R2:

```
interface FastEthernet5/0
 no ip address
 shutdown
!
router ospf 101
 log-adjacency-changes
 network 10.1.1.0 0.0.0.3 area 0
 network 10.2.2.0 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end
```

Figure 34 - Sauvegarde de la configuration R2

POUR R3:

```
interface Serial3/0
!
 ip address 10.2.2.1 255.255.255.252
 clock rate 64000
!
interface FastEthernet4/0
 no ip address
 shutdown
!
interface FastEthernet5/0
 no ip address
 shutdown
!
router ospf 101
 log-adjacency-changes
 network 192.168.3.0 0.0.0.255 area 0
 network 10.2.2.0 0.0.0.3 area 0
!
ip classless
 ip route 192.168.1.0 255.255.255.0 10.1.1.1
 ip route 192.168.1.0 255.255.255.0 10.1.1.2
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
 exec-timeout 5 0
 logging synchronous
 login local
!
line aux 0
!
line vty 0 4
 exec-timeout 5 0
 login local
 transport input ssh
!
!
!
end
```

Figure 35 - Sauvegarde de la configuration R3

Part 2: Configure a Site-to-Site VPN with Cisco IOS

In Part 2 of this lab, you will configure an IPsec VPN tunnel between R1 and R3 that passes through R2. You will configure R1 and R3 using the Cisco IOS CLI. You will then review and test the resulting configuration.

Task 1: Configure IPsec VPN Settings on R1 and R3.

Step 1: Verify connectivity from the R1 LAN to the R3 LAN.

```
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time=39ms TTL=125
Reply from 192.168.3.3: bytes=32 time=49ms TTL=125
Reply from 192.168.3.3: bytes=32 time=13ms TTL=125
Reply from 192.168.3.3: bytes=32 time=38ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 49ms, Average = 34ms

C:\>
```

Figure 36 - Ping PC-A

Step 2: Enable IKE policies on R1 and R3.

IPsec is an open framework that allows for the exchange of security protocols as new technologies, and encryption algorithms as they are developed.

There are two central configuration elements in the implementation of an IPsec VPN:

- Implement Internet Key Exchange (IKE) parameters
 - Implement IPsec parameters
- a. Verify that IKE is supported and enabled.
- IKE Phase 1 defines the key exchange method used to pass and validate IKE policies between peers. In IKE Phase 2, the peers exchange and match IPsec policies for the authentication and encryption of data traffic.

IKE must be enabled for IPsec to function. IKE is enabled, by default, on IOS images with cryptographic feature sets. If it is disabled, you can enable it with the **crypto isakmp enable** command. Use this command to verify that the router IOS supports IKE and that it is enabled.

POUR R1:

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto isakmp enable
R1(config)#end
R1#
*Jan 25 22:17:15.839: %SYS-5-CONFIG_I: Configured from console by console
R1#show running-config | include crypto isakmp
```

POUR R3:

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#crypto isakmp enable
R3(config)#end
R3#
*Jan 25 22:26:34.859: %SYS-5-CONFIG_I: Configured from console by console
R3#show running-config | include crypto isakmp
R3#
```

Figure 37 - Enable IKE policies on R1 and R3.

- b. Establish an ISAKMP policy and view the available options.

To allow IKE Phase 1 negotiation, you must create an ISAKMP policy and configure a peer association involving that ISAKMP policy. An ISAKMP policy defines the authentication and encryption algorithms and the hash function used to send control traffic between the two VPN endpoints. When an ISAKMP security association has been accepted by the IKE peers, IKE Phase 1 has been completed. IKE Phase 2 parameters will be configured later.

Issue the **crypto isakmp policy number** global configuration mode command on R1 for policy 10.

POUR R1:

```
R1(config)# crypto isakmp policy 10
```

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#hash sha
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)# group 14
R1(config-isakmp)#lifetime 3600
R1(config-isakmp)#end
R1#
*Jan 25 22:19:11.275: %SYS-5-CONFIG_I: Configured from console by console
R1#show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm: AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #14 (2048 bit)
  lifetime:            3600 seconds, no volume limit
R1#
```


POUR R3:

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#crypto isakmp policy 10
R3(config-isakmp)# encryption aes 256
R3(config-isakmp)# hash sha
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)# group 14
R3(config-isakmp)# lifetime 3600
R3(config-isakmp)#end
R3#
*Jan 25 22:30:20.475: %SYS-5-CONFIG_I: Configured from console by console
R3#show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
  encryption algorithm: AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #14 (2048 bit)
  lifetime:            3600 seconds, no volume limit
R3#
```

Figure 38 - Écran CLI

- c. View the various IKE parameters available using Cisco IOS help by typing a question mark (?).

POUR R1: R1 (config-isakmp) # ?

```
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#?
ISAKMP commands:
 authentication Set authentication method for protection suite
 default         Set a command to its defaults
 encryption      Set encryption algorithm for protection suite
 exit           Exit from ISAKMP protection suite configuration mode
 group          Set the Diffie-Hellman group
 hash           Set hash algorithm for protection suite
 lifetime       Set lifetime for ISAKMP security association
 no             Negate a command or set its defaults
R1(config-isakmp)#
```

POUR R3:

```
R3(config-isakmp)#?
ISAKMP commands:
 authentication Set authentication method for protection suite
 default         Set a command to its defaults
 encryption      Set encryption algorithm for protection suite
 exit           Exit from ISAKMP protection suite configuration mode
 group          Set the Diffie-Hellman group
 hash           Set hash algorithm for protection suite
 lifetime       Set lifetime for ISAKMP security association
 no             Negate a command or set its defaults
R3(config-isakmp)#
```

Figure 39 - View the various IKE

Step 3: Configure the IKE Phase 1 ISAKMP policy on R1 and R3.

Your choice of an encryption algorithm determines how confidential the control channel between the endpoints is. The hash algorithm controls data integrity, ensuring that the data received from a peer has not been tampered with in transit. The authentication type ensures that the packet was sent and signed by the remote peer. The Diffie-Hellman group is used to create a secret key shared by the peers that has not been sent across the network.

- a. Configure an ISAKMP policy with a priority of **10**. Use **pre-shared key** as the authentication type, **aes 256** for the encryption algorithm, **sha** as the hash algorithm, and the Diffie-Hellman group **14** key exchange. Give the policy a lifetime of **3600** seconds (one hour).

Note: Older versions of Cisco IOS do not support AES 256 encryption and SHA as a hash algorithm. Substitute whatever encryption and hashing algorithm your router supports. Ensure that the same changes are made on R3 in order to be in sync.

```
R1#show crypto isakmp policy
```

```
Global IKE policy
Protection suite of priority 10
  encryption algorithm:    AES - Advanced Encryption Standard (256 bit keys)
  hash algorithm:         Secure Hash Standard
  authentication method:   Pre-Shared Key
  Diffie-Hellman group:    #2 (1024 bit)
  lifetime:                3600 seconds, no volume limit
```

- b. Configure the same policy on R3.

```
R3#show crypto isakmp policy
```

```
Global IKE policy
Protection suite of priority 10
  encryption algorithm:    AES - Advanced Encryption Standard (256 bit keys).
  hash algorithm:         Secure Hash Standard
  authentication method:   Pre-Shared Key
  Diffie-Hellman group:    #2(1024 bit)
  lifetime:                3600 seconds, no volume limit
```

```
R3#
```

Step 4: Configure pre-shared keys.

```
R1#show running-config | include crypto isakmp key
crypto isakmp key cisco123 address 10.2.2.1
R1#
```

R3 (vers R1) :

```
R3#show running-config | include crypto isakmp key
crypto isakmp key cisco123 address 10.1.1.1
R3#
```

Step 5: Configure the IPsec transform set and lifetime.

a. Voir les options

```
R1(config)#crypto ipsec transform-set 50 ?
ah-md5-hmac    AH-HMAC-MD5 transform
ah-sha-hmac     AH-HMAC-SHA transform
esp-3des        ESP transform using 3DES(EDE) cipher (168 bits)
esp-aes         ESP transform using AES cipher
esp-des         ESP transform using DES cipher (56 bits)
esp-md5-hmac    ESP transform using HMAC-MD5 auth
esp-sha-hmac    ESP transform using HMAC-SHA auth
```

b. Créer le transform-

POUR R1 :

```
R1#show crypto ipsec transform-set
Transform set 50: {    { esp-256-aes esp-sha-hmac  }
    will negotiate = { Tunnel,  },

Transform set #$/default_transform_set_1: { esp-aes esp-sha-hmac  }
    will negotiate = { Transport,  },
Transform set #$/default_transform_set_0: { esp-3des esp-sha-hmac  }
    will negotiate = { Transport,  },
```

PROUR R3 :

```
R3#show crypto ipsec transform-set
Transform set 50: {    { esp-256-aes esp-sha-hmac  }
    will negotiate = { Tunnel,  },

Transform set #$/default_transform_set_1: { esp-aes esp-sha-hmac  }
    will negotiate = { Transport,  },
Transform set #$/default_transform_set_0: { esp-3des esp-sha-hmac  }
    will negotiate = { Transport,  },
```

Quelle est la fonction du jeu de transformations (transform set) dans IPsec ?

le transform set IPsec précise les méthodes de chiffrement et de protection des données utilisées dans le tunnel VPN.

c. Changer la durée de vie

IPsec SA R1 :

```
R1#show running-config | include security-association
crypto ipsec security-association lifetime seconds 1800
R1#
```

R3 :

```
R3#show running-config | include security-association
crypto ipsec security-association lifetime seconds 1800
R3#
```

Step 6: Define interesting traffic.

```
R1#show access-lists
Extended IP access list 101
 10 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
--
```

R3 :

```
R3#show access-lists
Extended IP access list 101
 10 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
```

IPsec évalue-t-il si les listes de contrôle d'accès (ACL) sont symétriques avant de négocier son association de sécurité ?

IPsec checks that the access control lists (ACLs) are symmetric (mirrored) on both devices to successfully negotiate a Security Association (SA).

Step 7: Create and apply a crypto map.

a. Créer la crypto map (R1)

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto map CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
      and a valid access list have been configured.
R1(config-crypto-map)#
```

b. Associer l'ACL

```
R1#show crypto map
Crypto Map CMAP 10 ipsec-isakmp
  WARNING: This crypto map is in an incomplete state!
    (missing peer or access-list definitions)
  Extended IP access list 101
    access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
  Security association lifetime: 4608000 kilobytes/1800 seconds
  PFS (Y/N): N
  Transform sets={
  }
  Interfaces using crypto map CMAP:
```

c. Voir les options

```
R1(config-crypto-map)#set ?
  peer           Allowed Encryption/Decryption peer.
  pfs            Specify pfs settings
  security-association Security association parameters
  transform-set   Specify list of transform sets in priority order
R1(config-crypto-map)#set
% Incomplete command.
R1(config-crypto-map)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show crypto map
Crypto Map CMAP 10 ipsec-isakmp
  WARNING: This crypto map is in an incomplete state!
    (missing peer or access-list definitions)
  Extended IP access list 101
    access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
  Security association lifetime: 4608000 kilobytes/1800 seconds
  PFS (Y/N): N
  Transform sets={
  }
  Interfaces using crypto map CMAP:
```

d. Définir le peer

```
R1#show crypto map
Crypto Map CMAP 10 ipsec-isakmp
  Peer = 10.2.2.1
  Extended IP access list 101
    access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
  Current peer: 10.2.2.1
  Security association lifetime: 4608000 kilobytes/1800 seconds
  PFS (Y/N): N
  Transform sets={
  }
  Interfaces using crypto map CMAP:
```

e. Finaliser la crypto map (R1)

```
R1(config-crypto-map)#set pfs group2
R1(config-crypto-map)#set transform-set 50
R1(config-crypto-map)#set security-association lifetime seconds 900
R1(config-crypto-map)#exit
R1(config)#
```

f. Crypto map miroir sur R3

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#crypto map CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R3(config-crypto-map)#match address 101
R3(config-crypto-map)#set peer 10.1.1.1
R3(config-crypto-map)#set pfs group2
R3(config-crypto-map)#set transform-set 50
R3(config-crypto-map)#set security-association lifetime seconds 900
R3(config-crypto-map)#exit
```

g. Appliquer sur les interfaces

WAN R1 :

```
R1#show crypto map
Crypto Map CMAP 10 ipsec-isakmp
  Peer = 10.2.2.1
  Extended IP access list 101
    access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
  Current peer: 10.2.2.1
  Security association lifetime: 4608000 kilobytes/900 seconds
  PFS (Y/N): Y
  Transform sets={
    50,
  }
  Interfaces using crypto map CMAP:
    Serial0/0/0
```

R3 :

```
R3#show crypto map
Crypto Map CMAP 10 ipsec-isakmp
  Peer = 10.1.1.1
  Extended IP access list 101
    access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
  Current peer: 10.1.1.1
  Security association lifetime: 4608000 kilobytes/900 seconds
  PFS (Y/N): Y
  Transform sets={
    50,
  }
  Interfaces using crypto map CMAP:
    Serial0/0/1
```

Task 2: Verify the Site-to-Site IPsec VPN Configuration.

Step 1: Verify the IPsec configuration on R1 and R3.

a. Vérification des politiques ISAKMP et des transform

sets IPsec R1 :

```
R1#show crypto ipsec transform-set
Transform set 50: {   { esp-256-aes esp-sha-hmac  }
  will negotiate = { Tunnel,  },

Transform set #$/default_transform_set_1: { esp-aes esp-sha-hmac  }
  will negotiate = { Transport,  },
Transform set #$/default_transform_set_0: { esp-3des esp-sha-hmac  }
  will negotiate = { Transport,  },
```

R3 :

```
R3#show crypto ipsec transform-set
Transform set 50: {   { esp-256-aes esp-sha-hmac  }
  will negotiate = { Tunnel,  },

Transform set #$/default_transform_set_1: { esp-aes esp-sha-hmac  }
  will negotiate = { Transport,  },
Transform set #$/default_transform_set_0: { esp-3des esp-sha-hmac  }
  will negotiate = { Transport,  },
```

b. Vérification des crypto maps appliquées sur le

routeur R1 :

```
R1#show crypto map
Crypto Map CMAP 10 ipsec-isakmp
  Peer = 10.2.2.1
  Extended IP access list 101
    access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
  Current peer: 10.2.2.1
  Security association lifetime: 4608000 kilobytes/900 seconds
  PFS (Y/N): Y
  Transform sets={
    50,
  }
  Interfaces using crypto map CMAP:
    Serial0/0/0
```

R3:

```
R3#show crypto map
Crypto Map CMAP 10 ipsec-isakmp
  Peer = 10.1.1.1
  Extended IP access list 101
    access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
  Current peer: 10.1.1.1
  Security association lifetime: 4608000 kilobytes/900 seconds
  PFS (Y/N): Y
  Transform sets={
    50,
  }
  Interfaces using crypto map CMAP:
    Serial0/0/1
```

Task 3: Verify the IPsec VPN Operation.

Step 1: Display ISAKMP security associations.

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
IPv6 Crypto ISAKMP SA
```


Step 2: Display IPsec security associations.

```
R1#show crypto ipsec sa

interface: Serial0/0/0
  Crypto map tag: CMAP, local addr 10.1.1.1

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
  current_peer 10.2.2.1 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 10.1.1.1, remote crypto endpt.:10.2.2.1
    path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
    current outbound spi: 0x0(0)

  inbound esp sas:
```

Pourquoi aucune association de sécurité (SA) n'a-t-elle été négociée ?

Pas de SA négociée : le trafic ne déclenche pas la création d'un tunnel IPsec.

Step 3: Generate some uninteresting test traffic and observe the results.

a. Test de connectivité WAN entre R1 et R3

```
R1#ping 10.2.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/3/7 ms

R1#
```

b. Vérification des associations de sécurité ISAKMP

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status

IPv6 Crypto ISAKMP SA

R1#
```

c. Test de connectivité LAN entre R1 et R3

```
R1#ping 192.168.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/13/55 ms
```

d. Nouvelle vérification des associations ISAKMP

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status

IPv6 Crypto ISAKMP SA
```

e. Observation des paquets OSPF Hello

```
R1#debug ip ospf events
OSPF events debugging is on
R1#
02:10:45: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/0/0 10.1.1.2

02:10:45: OSPF: End of hello processing

R1#
02:10:55: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/0/0 10.1.1.2

02:10:55: OSPF: End of hello processing

R1#
02:11:05: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/0/0 10.1.1.2

02:11:05: OSPF: End of hello processing

R1#
02:11:15: OSPF: Rcv hello from 10.2.2.2 area 0 from Serial0/0/0 10.1.1.2

02:11:15: OSPF: End of hello processing
```

f. Désactivation du débogage OSPF

```
R1#undebug all
All possible debugging has been turned off
R1#
```

f. Vérification finale des associations ISAKMP

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status

IPv6 Crypto ISAKMP SA
```

Step 4: Generate some interesting test traffic and observe the results.

a. Ping étendu de R1 vers R3 G0/1

```
R1#ping
Protocol [ip]:
Target IP address: 192.168.3.1
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: 192.168.1.1
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/4/5 ms
```

b. Vérifier les associations de sécurité ISAKMP après le ping

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.2.2.1     10.1.1.1     QM_IDLE       1071      0 ACTIVE

IPv6 Crypto ISAKMP SA
```

c. Vérifier les associations de sécurité IPsec et les paquets transformés

```
R1#ping 192.168.3.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.3, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/8/15 ms
```

```
R1#show crypto ipsec sa

interface: Serial10/0/0
  Crypto map tag: CMAP, local addr 10.1.1.1

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
  remote  ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
  current_peer 10.2.2.1 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 4, #pkts encrypt: 4, #pkts digest: 0
    #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

    local crypto endpt.: 10.1.1.1, remote crypto endpt.:10.2.2.1
    path mtu 1500, ip mtu 1500, ip mtu idb Serial10/0/0
    current outbound spi: 0xDE38338B(3728225163)

  inbound esp sas:
    spi: 0x6825DB7E(1747311486)
```

d. Autres types de trafic pouvant déclencher un SA

Seul le trafic correspondant aux ACL définies dans la crypto map, comme HTTP/HTTPS, FTP, Telnet, SSH ou OSPF/RIP sur des interfaces sécurisées, déclenche la création d'un tunnel IPsec.

Réponses réflexion

- Le trafic entre **PC-A** et l'interface **G0/0** de **R1** n'est **pas chiffré**.
- Le chiffrement IPsec ne commence qu'**après R1**, sur l'interface WAN où la **crypto map** est appliquée.
- Les types de trafic qui peuvent déclencher le tunnel VPN sont : **HTTP**, **FTP**, **SSH**, **ICMP**, ou tout autre trafic IP correspondant à l'**ACL 101**.

